



Store at: +2+8°C.

Presentation:

Cod. EZ021LQ

EZ021LQ-SP

EZ022LQ

CONT: R1 1 x 100 R2 1 x 25 mL.

CONT: R1 1 x 40 R2 1 x 10 mL.

CONT: R1 2 x 100 R2 2 x 25 mL.

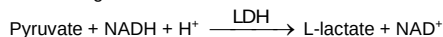
Procedure

Quantitative determination of lactate dehydrogenase (LDH).

Only for in vitro use in clinical laboratory (IVD)

TEST SUMMARY

Lactate dehydrogenase (LDH) catalyses the reduction of pyruvate by NADH, according the following reaction:



The rate of decrease in concentration of NADPH, measured photometrically, is proportional to the catalytic concentration of LDH present in the sample¹.

REAGENTS COMPOSITION

R 1	Imidazole	65 mmol/L
Buffer	Pyruvate	0.6 mmol/L
R 2	NADH	0.18 mmol/L
Substrate		

REAGENT PREPARATION AND STABILITY

Working reagent (WR):

Mix 4 volumes of R1 with 1 volume of R2.

Stability: 15 days at 2-8°C or 5 days at room temperature (15-25°C).

Signs of Reagent deterioration:

- Presence of particles and turbidity.
- Blank absorbance (A) at 340 nm. < 1.00

All the components of the kit are stable until the expiration date on the label when stored at 2-8°C, protected from light and contamination prevented during their use. Do not use reagents over the expiration date.

SPECIMEN

Serum¹. Separated from cells as rapidly as possible. Do not use oxalates as anticoagulants since they inhibit the enzyme.

Do not use haemolysed samples.

Stability: 2 days at 2-8°C.

MATERIAL REQUIRED BUT NOT PROVIDED

- Spectrophotometer or colorimeter measuring at 340 nm.
- Thermostatic bath at 25°C, 30°C or 37°C (± 0.1°C)
- Matched cuvettes 1.0 cm light path.

General laboratory equipment.

TEST PROCEDURE

- Assay Conditions
 - Wavelength : 340 nm.
 - Cuvette: 1 cm light path.
 - Constant temperature 25°C / 30°C / 37°C.
- Adjust the instrument to zero with distilled water or air.
- Pipette into a Cuvette:

	25°C. - 30°C	37°C.
WR (mL)	3.0	3.0
Sample (μL)	100	50

- Mix. Incubate for 1 minute.
- Read the absorbance (A) of the sample, start the stopwatch and read absorbance at 1 min. interval thereafter for 3 min.
- Calculate the difference of absorbance and the average absorbance difference per minute (ΔA/min.).

CALCULATIONS^(Note 2)

$$\begin{aligned} 25^\circ - 30^\circ\text{C} & \quad \Delta A/\text{min} \times 4925^* = \text{U/L LDH} \\ 37^\circ\text{C} & \quad \Delta A/\text{min} \times 9690^* = \text{U/L LDH} \end{aligned}$$

Units: One international unit (IU) is the amount of enzyme that transforms 1 μmol of substrate per minute, in standard conditions. The concentration is expressed in units per litre of sample (U/L).

Temperature conversion factors

To correct results to other temperatures multiply by:

Assay temperature	Conversion factor to		
	25°C	30°C	37°C
25°C	1.00	1.33	1.92
30°C	0.75	1.00	1.43
37°C	0.52	0.70	1.00

QUALITY CONTROL

Control sera are recommended to monitor the performance of the procedure, H Normal and H Pathological (QC003, QC004).

If control values are found outside the defined range, check the instrument, reagents and calibrator for problems.

Serum controls are recommended for internal quality control. Each laboratory should establish its own Quality Control scheme and corrective actions if controls do not meet the acceptable tolerances.

REFERENCE VALUES¹

25°C	30°C	37°C
120-240 U/L	160-320 U/L	230-460 U/L

(These values are for orientation purpose).

It is suggested that each laboratory establish its own reference range.

CLINICAL SIGNIFICANCE

Lactate dehydrogenase (LDH) is an enzyme with wide tissue distribution in the body.

The higher concentrations of LDH are found in liver, heart, kidney, skeletal muscle and erythrocytes.

Increased levels of the enzyme are found in serum in liver disease, myocardial infarction, renal disease, muscular dystrophy and anemia^{1,4,5}.

Clinical diagnosis should not be made on a single test result; it should integrate clinical and other laboratory data.

REAGENT PERFORMANCE

- Measuring Range:

From detection limit of 3.42 U/L to linearity limit of 1700 U/L, under the described assay conditions.

If results obtained were greater than linearity limit, dilute the sample 1/10 with NaCl 9 g/L. and multiply result by 10.

- Precision:

	Intra-assay n= 20		Inter-assay n= 20	
Mean (U/L)	400	785	392	773
SD	3.15	10.97	6.23	9.93
CV	0.79	1.40	1.59	1.26

- Sensitivity: 1 U/L = 0.00009 ΔA/min

- **Accuracy:** Results obtained GPL reagents did not show systematic differences when compared with other commercial reagents.

The results obtained using 100 samples were the following:

Correlation coefficient (r)²: 0.98382

Regression Equation: y= 0.8988x + 2.583

The results of the performance characteristics depend on the analyzer used.

INTERFERING SUBSTANCES

- Haemolysis interferes with the assay.

- Some anticoagulants such as oxalates interfere with the reaction¹.

- A list of drugs and other interfering substances with LDH determination has been reported by Young et. al^{2,3}

NOTES

- Use clean disposable pipette tips for its dispensation.
- Formulation to reach constant:

$$\begin{aligned} \Delta A/\text{Min.} \times 4925^* & \text{ or } \\ 9690^* & = \text{U/L LDH} \end{aligned}$$

* $\frac{\text{Tv} \times 1000}{\epsilon \times \text{LP} \times \text{Sv}}$	Tv= Total volume in mL ε NAHD = 6.22 at 340 nm LP= Light path Sv= Sample volume in mL
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BIBLIOGRAPHY

- Pesce A. Lactate dehydrogenase. Kaplan A et al. Clin Chem The C.V. Mosby Co. St Louis. Toronto. Princeton 1984; 1124-117, 438.
- Young DS. Effects of drugs on Clinical Lab. Tests, 4th ed AACC Press, 1995.
- Young DS. Effects of disease on Clinical Lab. Tests, 4th ed AACC 2001.
- Burtis A et al. Tietz Textbook of Clinical Chemistry, 3rd ed AACC 1999.
- Tietz N W et al. Clinical Guide to Laboratory Tests, 3rd ed AACC 1995.



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